Nikola Tesla Receives Edison Medal

Notable Program at Annual Meeting of American Institute of Electrical Engineers—Interesting Insight into Mr. Tesla's Life and Work as Disclosed in His Address of Acceptance

Nikola Tesla, inventor and scientist, was presented the seventh annual Edison Medal at the annual meeting of the American Institute of Electrical Engineers held at New York City on May 18. Announcement of this award was made in our issue of May 5. This notable event, with the memorable addresses made by A. E. E. Kennelly, Charles A. Terry, B. A. Behrend, and particularly by Mr. Tesla himself, formed the principal item of the evening's program, which also included the annual report of the Board of Directors and announcement of letter ballot on the annual election of officers, which are reported elsewhere in this issue. After these business matter, had been disposed of, President Buck called on A. E. Kennelly, chairman of the Edison Medal Committee.

Nikola Tesla.

Dr. Kennelly gave an account of how the Edison Medal originated in a group of Mr. Edison's friends setting apart a fund in 1904 for the annual award to a living electrician of a medal for "meritorious achievement in electrical science and art." A deed of gift was drawn up designating the American Institute of Electrical Engineers to appoint from its officers and members a committee of 24 to determine the awards. Six noted electrical men have been recipients of the Edison Medal, as follows: Elihu Thomson, Frank J. Sprague, George Westinghouse, William Stanley, Charles F. Brush and Alexander Graham Bell. Dr. Kennelly stated that the medal award to Mr. Tesla was particularly for his development of the rotating magnetic field which is the fundamental principle that made possible the use of alternating current for operating motors, and also for his investigations of the phenomena of high-frequency currents.

Mr. Terry's Address.

Charles A. Terry, who was associated with Mr. Tesla in some of his research work, then made an eloquent address on Mr. Tesla's electrical achievements. After dwelling on the unselfish creative impulse of the inventor, Mr. Terry said: "Twenty-nine years ago this month there was presented before this Institute a paper of unusual import. It was entitled 'A New System of Alternate-Current Motors and Transformers.' The author, Nikola Tesla, was then only 31 years of age, and but four years a resident of this country. His early life was spent near his birthplace not far from the Eastern Adriatic Coast. His father, a Greek clergyman, and his mother, herself of an inventive mind, secured for their young son a comprehensive training in mathematics, physics and philosophy. At the age of 22 he had completed his studies in engineering at the Polytechnic School in Gratz and also a course in the University of Prague; and in 1881 began his practical work at Budapest. In 1883 he was located at Strasburg, engaged in completing the lighting of a newly erected railway station. Shortly after finishing this task he came to the United States. Mr. Tesla's first work in this country was upon new designs of direct-current arc and incandescent lighting systems for the Edison Company.

"Throughout all these years his desire had been to find an opportunity to demonstrate the truth of a conviction which became fixed in his mind while studying direct-current motors in school at Gratz in 1878; the conviction was that it should be possible to create a rotating magnetic field without the use of commutators. * * *

"By some fortunate process of reasoning he conceived, while in Budapest in 1882, that by using two or more out-of-phase alternating currents respectively passing through geometrically displaced coils it would be possible to develop his long sought progressively shifting magnetic field.

"Lack of funds and facilities for working out his theory compelled still further postponement, but in 1885 Tesla had the good fortune to interest men of means in a direct-current arc light which he had devised, and subsequently a laboratory was equipped for him in Liberty Street, New York, and here at last he found opportunity to demonstrate the correctness of his long cherished theory. In 1887 he was able to exhibit to his business associates and to Prof. William A. Anthony, whose expert opinion they sought, motors having such proggressively shifting fields without the use of commutators, as he had foreseen nine years before.

"Having thus demonstrated the correctness of his theory and the feasibility of its application, it remained for Tesla to work out various practical methods of applying the principle, and the rapidity and wonderful way in which he surrounded the entire field of constant-speed synchronous, induction and split-phase motors is beautifully set forth in his paper of May 18, 1888, and in the numerous patents issued May 1, 1888, and succeeding years, covering the forms of electric motors which have since become the almost universal means for transforming the energy of alternating currents into mechanical energy. * * *

"Among the first to recognize the immense importance of Mr. Tesla's motors were Mr. Westinghouse and his advisors, Mr. Kerr, Mr. Byllesby, Mr. Shallenberger and Mr. Schmid, and in June Mr. Westinghouse secured an option which shortly resulted in the purchase of the patents, thus bringing under one ownership the alternating-current transformer system of distribution, and the Tesla motor. * * *

"The impress made upon the world by the deeds of a great inventor can not be measured by the number of patents which he has received nor by the monetary reward secured nor by the mere exploitation of his name. Often his greatest gifts are in the form of inspiring contributions to the literature, filled with suggestions of lines of thought which lead others to work in untried fields. This is especially true of a series of lectures delivered by Mr. Tesla upon the subject of high-frequency, high-potential currents. The first of the series was given at Columbia College in 1891, before this Institute. During 1892 and 1893 this lecture with additional data and experiments was repeated in London, Paris, Philadelphia and St. Louis. * * * During these lectures he exhibited to the audience numerous experiments displaying striking and instructive phenomena. He also described many novel pieces of apparatus, such, for instance, as his high-frequency generator and induction coils and his magnetically quenched arc. * * *"

Mr. Terry quoted from many noted contemporaries of Mr. Tesla at that time their appreciation of the pioneer nature and

importance of his investigations in high-frequency phenomena and wireless transmission. As early as February, 1892, he had predicted these developments and "in his lecture before the National Electric Light Association at St. Louis in March, 1893, Mr. Tesla elaborated certain views regarding the importance of resonance effects in this field. * * * He then announced that his conviction had grown so strong that he no longer looked upon the plan of transmitting intelligence as a mere theoretical possibility, and referring to the existing belief of some that telephony to any distance might be accomplished 'by induction through the air,' concisely set forth his theory as follows: 'I cannot stretch my imagination so far, but I do firmly believe that it is practical to disturb by means of powerful machines the electrostatic condition of the earth and thus transmit intelligible signals and perhaps power.' * * *

"Throughout Tesla's work with high-potential currents he had persistently in mind the wireless transmission of power in large quantities. It was in the furtherance of this line of investigation that he expended large amounts of money and years of labor at Wardenclyffe, Long Island, and at Telluride, Colo. Late in 1914 he secured a patent upon an application filed twelve years before upon an apparatus for transmitting electric energy with which he hopes to be able to transmit unlimited power with high economy to any distance without wires. While as yet these efforts have not resulted in commercial exploitation, the future may prove that his dream of thus transmitting energy in substantial amounts is of that class which in time comes true, as in the case of his dream of wireless telegraphy.

"Another use to which Tesla adapted the results of his high-frequency investigations was the control of the movements of torpedoes and boats. In 1898 he patented such an apparatus and also built and successfully operated such a craft. The movements of the propelling engine, the steering and other mechanisms were controlled wirelessly from the shore or other point through a distance of two miles. Apparently this, like some of his other inventions, was ahead of its time. * * *

"Another field of investigation, in which Mr. Tesla has contributed valuable material is related to the Roentgen ray. In the Electrical Review during March and April, 1896, there appeared a number of communications from Mr. Tesla which, while giving full credit to Roentgen for his magnificent discovery, make public much additional data derived from his own careful experiments in this line of research. * * *

"Among the many other inventions to which Mr. Tesla has devoted much time and energy may be mentioned a thermomagnetic motor and a pyromagnetic generator, anti-sparking dynamo brush and commutator, auxiliary brush regulation of direct-current dynamos, unipolar dynamos, mechanical and electrical oscillators, electrotherapeutic apparatus, the oxidation of nitrogen by high-frequency currents, and an electrolytic registering meter. * * *

"One other line of endeavor entirely outside of electricity to which Tesla has given much attention is the development of a bladeless steam turbine in which the friction of the passing steam as distinguished from its direct impact is availed of. The steam is admitted between plain parallel rotating discs and passing spirally from the circumference toward the axial center imparts energy to the disks. Such a turbine can be run at exceedingly high temperatures, is readily reversible and, having no blades, is extremely simple and free from liability to accidental derangement. With great ingenuity Tesla has succeeded in producing such machines of considerable power and having exceedingly interesting characteristics. It is to be hoped that with his indefatigable zeal Tesla will soon succeed in perfecting the commercial application of this invention. It is not possible in this brief survey even to touch upon many of the lines of Mr. Tesla's varied activities, but we must content ourselves with this inadequate presentation of typical evidences of the fascinating genius of this man."

Address by B. A. Behrend.

"By an extraordinary coincidence, it is exactly 29 ago, to the very day and hour, that there stood before Institute Nikola Tesla," said Mr. Behrend. He then q from Mr. Tesla's paper on that notable occasion. Contin he said:

"Not since the appearance of Faraday's experiment searches in electricity has a great experimental truth voiced so simply and so clearly as this description of Tesla's great discovery of the generation and utilizatic polyphase alternating currents. He left nothing to be for those who followed him. His paper contained the ske even of the mathematical theory.

"Three years later, in 1891, there was given the first demonstration, by Swiss engineers, of the transmissic power at 30,000 volts from Lauffen to Frankfort by mea Mr. Tesla's system. A few years later this was follow the development of the Cataract Construction Company, the presidency of our member, Edward A. Adams, and the aid of the engineers of the Westinghouse Company. interesting to recall here tonight that in Lord Kelvin's r to Mr. Adams, Lord Kelvin recommended the use of current for the development of power at Niagara Falls for its transmission to Buffalo.

"The due appreciation or even enumeration of the r of Mr. Tesla's invention is neither practicable nor desira this moment. There is a time for all things. Suffice it t that, were we to seize and to eliminate from our indu world the results of Mr. Tesla's work, the wheels of ind would cease to turn, our electric cars and trains would our towns would be dark, our mills would be dead and Yea, so far-reaching is this work that it has become the and woof of industry. * * *"

Presentation of the Medal.

President H. W. Buck, in presenting the medal tc Tesla, referred to the developments of some 30 years ago direct currents were used almost exclusively and it rem for Mr. Tesla to make alternating currents available for cellaneous purposes by his great conception of the rot field. This made power transmission immediately practi and permitted the development of the Niagara Falls ente and countless other power developments necessitating th of polyphase currents. Mr. Buck also spoke of Tesla's pi work in high-voltage and high-frequency currents, which the foundation for such important later developments.

Mr. Tesla's Address of Acceptance.

In accepting the Edison Medal, Mr. Tesla expresse deep gratitude for the sympathy and appreciation shown He admitted that "a gigantic revolution has been wroug the transmission and transformation of energy. While w pleased with the results achieved," he said, "we are pre on, inspired with the hope and conviction that this is jus beginning, a forerunner of further and still greater ac plishments." He then entered upon a fascinating accou some features of a personal and more intimate character ing on his work, which lack of space unfortunately pre giving in full.

"In the first place, I come from a very wiry and long-race. Some of my ancestors have been centenarians, anc of them lived 129 years. I am determined to keep up the ord, and believe there is a prospect of accomplishing it. nature has given me a vivid imagination which, throug cessant exercise and training, through the study of scie subjects, and the verification of theories through experi has become very accurate in results, so that I have been to dispense, to a large extent, with the slow labors, was and expensive processes of practical development of the I conceive. It has made it possible for me to explore exte

fields with great rapidity and get results with the least expenditure of vital energy. By this means, I may tell you also, I am able to picture the objects of my desires in forms so real and tangible that I can rid myself of that morbid craving for perishable possessions to which so many succumb. * * *

"My life was also wonderful in another respect, for physical endurance or energy. If you inquire into the career of successful men in the inventor's profession, you will find, as a rule, that they are as remarkable for their physical as for their mental capacities. I know that when I worked with Edison, after all of his assistants had been exhausted, he said to me, 'I never saw such a thing, you take the cake.' That was the characteristic way for Edison to express what I did. We worked from in the morning at half past ten until five o'clock the next morning. I carried this on for nine months without a single day's exception, but everybody else gave up. Edison stuck, but he occasionally dozed off on the table. * * *

"When I turned my thoughts to inventions, I found that I could visualize my conceptions with the greatest facility. I did not need any models and drawings or experiments, I could do it all in my mind, and I did. The way I unconsciously evolved what I considered a new method in materializing inventive concepts and ideas, is exactly opposed to the purely experimental method, of which undoubtedly Edison is the greatest and most successful exponent. The moment you construct a device to carry into practice a crude idea you will find that you will be engrossed with the details and effects of the apparatus. As you go on changing and constructing, you will lose the forces of concentration, and you will lose sight of the great underlying principle. You obtain results, but at the sacrifice of quality. I did not construct. When I got an idea, I started right away to build it up in my mind. I changed the structure, I made improvements, I experimented, and I ran the device in my mind. It is absolutely the same to me whether I place my turbine in my mind or have it in my shop actually running in my test. It makes no difference. The results are the same. In this way you see I can develop and perfect an invention without touching anything, and when I have gone so far that I have put into that device every possible improvement I can think of, that I can see no fault in it any more, I then construct it, and every time my device works as I conceived it would, my experiment comes out exactly as I plan it, and in twenty years there has not been a single, solitary experiment which did not come out exactly as I thought it would."

Mr. Tesla then related some very interesting experiences of his youth in which he had hairbreadth and almost miraculous escapes that led his parents to set him apart for the ministry. At the age of 18 during a violent epidemic of cholera he contracted the disease with several complications that made everyone despair of his recovery. "My father came, sat on my bed and said to me, 'Are you going to get well?' I said to him, 'I will get well if you will let me study engineering.' He said, 'Certainly I will, you will have to go to the best polytechnic school in Europe.' My father kept his word, and after a year of roaming through the mountains and getting myself in good physical shape, I went to the Polytechnic School at Gratz, one of the finest in Europe. * * *

"The first year at the polytechnic school was passed in this way—I got up at three o'clock in the morning and I worked until eleven o'clock at night, for one whole year, without a single day's exception. Well, you know when a man with a reasonably healthy brain works that way he must do something. Of course, I did. I graduated nine times that year and some of the professors were not satisfied with giving me the highest distinction, because they said that did not express their idea of me, and here is where I come to the rotating field. In addition to the regular graduating papers they gave me some certificates. I got these certificates and showed them to my father. I thought it would be a great triumph for him. My father took the certificates and threw them into the waste basket. He said, 'Oh, I know how these certificates are obtained.' That almost killed my ambition, and later, after my father died, I was mortified to find a pack of letters, from which I could see that there had been considerable correspondence going on between my father and the professors at the school, and the professors had written to the effect that unless he took me away from the school I would kill myself with work. Then I understood why he had slighted my success, which I was told was greater than any which had been achieved by any student before; in fact, the best students had only graduated twice, and I had graduated nine times. My work in the first year had this result—that the professors became very much interested in me and very much attached to me.

"It was in the second year of my studies that we got a gramme machine there from Paris. This was a machine with a horseshoe form of laminated magnets, and there was an armature in it with a commutator. I put that together and we connected it up and showed the various effects of currents. During this time that Professor Poeschl was demonstrating we had some trouble with the brushes. The brushes sparked very badly, and I said, 'Well, why should not we operate without the brushes?' Professor Poeschl said it could not be done, but in view of my success in the past year he gave me the honor of delivering a lecture. He said: 'Mr. Tesla may accomplish great things, but he certainly never can produce this,' and he showed that this would be tantamount to converting a steadily pulling force, like gravity, into a rotary effort, which was a sort of perpetual-motion machine, an impossible idea. * * * I showed them that I was right and it was possible. It was not a perpetual-motion idea. It was not a thing which was impossible, it could be done, and I started then to work.

"I will not tire you with an extended account of this undertaking, but I will say that my work was like this—I began in 1877 in the summer, and then I would get a picture in my mind. I would picture, for instance, a direct-current machine and run it and see how these currents changed in the armature, and whatever I imagined, I would put together and operate everything in my mind, and I continued that practice until 1882. I could not visualize just exactly how to do it, but I felt I was approaching a solution. I was on my vacation and sure enough the idea came to me. I will never forget the moment. I was walking with a friend of mine in the city park at Budapest, and I was reciting poems from Faust. It was nothing for me to recite from memory the contents of an entire book, with every word between the covers, from the first to the last, and my sister and brother had a much better memory than mine. Just as the sun was setting I felt wonderfully inspired, and the idea came to me like a flash, and I saw the whole thing clearly, I saw my generator, I saw the motor, and saw the connections, saw it work, in my mind, and I took a stick and drew on the sand the diagrams which were drawn in my paper before the American Institute of Electrical Engineers, and which were illustrated in my plans as clearly as possible, and from that time I carried this image in my mind. Now, remember, if I had been a man who had had some of the practical gifts of Edison, I would have gone right away to perform an experiment and push the invention along, but I did not need to experiment—as I say, I can so picture, and consider what I picture as real and forcible, that I do not need any experimenting, nor is it particularly interesting to me. I went on and improved the plan continuously, invented new forms, and the day before I came to America, every form, every construction, every arrangement which I expressed in my patents, was developed in my mind, except just two or three forms which appear in later developments, otherwise I had all the forms shown in the thirty or forty patents granted to me then in my mind. In 1882 I performed two experiments in Strasburg, as Mr. Terry pointed out, and there at the railroad station I obtained the first rotation.

"I have come now to an interesting point when I came to America. I had made some improvements in dynamos for a French company who were getting their machinery from America. The improved forms were so much better that the manager of the works said to me, 'You must go to America, and you must design these machines for the French company.' So, after ineffectual efforts on the other side to get somebody to interest himself in my plans financially, I came to America. * * * Then came another event. I met Edison, and the effect that Edison produced upon me was rather extraordinary. When I saw this wonderful man, who had had no training at all, no advantages, and did it all himself, and see the great results by virtue of his industry and application, I felt mortified that I had squandered my life—you see I had studied a dozen languages, I had delved in literature, and had spent the best years of my life in ruminating through libraries and reading all sorts of stuff that fell into my hands. I thought to myself, what a terrible thing it was to have wasted my life on those useless things, and if I had only come to America right then and there and devoted all of my brain power and inventiveness to my work, what could I have not done? In later life I realized I might not have produced anything without the scientific training I got, and it is a question as to whether my theory as to my possible accomplishments was correct. In Edison's works we passed a year of the most strenuous labor, and then certain people approached me with a proposition to form my own company.

"Well, I went into the proposition, and we developed an arc light, and to show you how prejudiced the people were against the alternating current, as one of the speakers indicated, I told these friends of mine I had a great invention relating to alternating-current transmission, and they said: 'No, we want the arc lamp. We don't want this alternating-current system.' Finally I brought out an arc lamp, perfected it, and the city adopted it. Then I succeeded in organizing another company, in April, 1886, and then a laboratory was put up, and then I rapidly developed these matters, and finally the Westinghouse people approached us, and an arrangement was made, and then you know what happened since then. The invention has swept the world.

"Mr. Behrend has referred to other inventions of mine. I will say just a few words, because some of my work has been misunderstood. It seems to me that I ought to tell you a few words about other efforts that have absorbed my attention later. In 1894 I delivered a lecture at the Royal Institution and Lord Raleigh surprised me by acknowledging my work in very generous terms, something that is not customary, and among other things he stated that I had really an extraordinary gift for invention. Up to that time, I can assure you, I never realized that I was an inventor at all. I looked upon the rotating-field discovery as simply a mathematical deduction. It was a logical, step by step, deduction. I forced this invention, as it were, by screws and levers. I did not get an inspiration, as it were. My machines were developed all in my mind. When I tried the first experiments they meant nothing to me. I had already demonstrated them perfectly. So when I went home, in 1892, and heard these remarks of Lord Raleigh, I began to think and said to myself, 'Why, certainly I must have been an inventor.'

"I remember that I constructed turbines and clocks and such things, and so I said to myself, 'If I really have a gift for invention, then I am going to bend it to some great purpose and some great task and not squander my efforts in small things.' I began to think just what was the greatest thing to accomplish. One day I was working in the forest and a storm came up. I ran under a tree for shelter, and the air was very heavy, and all at once there was a lightning flash, and immediately after the lightning flash a torrent of rain came. That gave me a fair start. I realized that the sun was lifting the water and vapor, and the wind swept it over the regions where it accumulated and reached the condition where at a certain point it was condensed and fell to the earth again, this life-sustaining stream of water was entirely domin by some other power, and lightning or some other agenc this kind, simply came in as a trigger mechanism to relieve energy at the proper moment. I started out and attacked problem of constructing some kind of a machine which w this kind simply came in as a trigger mechanism to relieve enable us to precipitate this water wherever desired.* *

THE COLORADO EXPERIMENTS.

"That led me to the production of very intense elect effects. At the same time my wireless work, which I already begun, was exactly in that direction, and so I dev myself to the perfection of some device, and in 1908, I an application describing an apparatus with which I think wonder can be achieved. The Patent Office examiner from Missouri, and he would not believe it could be done, so my patent was never granted, but in Colorado I constru apparatus by which I produced certain effects, at least, w were greater than those of lightning—I do not mean, for stance, in potential. The highest potential I reached something like 20,000,000 volts, and that is insignifican compared to lightning, but certain effects in my apparatus greater than those produced by lightning. For instance, I in my attenna currents of 1000 or 1100 amperes, and you k that in the biggest wireless plants only 250 amperes were —I do not know what is the biggest one now—but 250 amp is very large, and I know in Colorado I succeeded one da precipitating a mist. Of course, there was a mist outside, when I turned on the current the mist in the laboratory came so great that I called the attention of my assistant when the hand was held just a few inches from the fac could not be seen, so that I am positive of my conviction we are able to erect a plant of proper design in an arid reg that we can work this plant according to certain observat and rules, and that we can draw from the ocean unlim amounts of water for irrigation and power purposes. If not live to carry it out, somebody else will, but I am sure I am right.

"As to the transmission of power through space, th a matter which I have considered certain, absolutely cer years ago. Years ago I was in a position to erect a plan transmit wireless power to any distance without limita other than the physical limitations of the globe. In my sy it makes no difference what the distance is. The efficienc the transmission can be as high as 96 per cent, or 97 per and there is no loss except those losses which are inevit in the running of the machinery. And when there is no ceiver there is no loss anywhere. When the receiver is pu it draws energy. That is the opposite of the Hertz wave tem. In that system you have a plant of 1000 horsepo that is 1000 horsepower radiating all the time, but if I ha plant of 1000 horsepower under my system, this only cons the few horsepower necessary to maintain the electric vi tion, and the rest of the system is idle, exactly as the Ed lamps are idle when the lamps and dynamos are shut off.

"I have made advances along this line in later years w will contribute to the practical features of the system. cently we have obtained a patent on a transmitter with w it is practicable to transfer unlimited amounts of energ any distance. I had a very interesting experience with Stone, whom I consider, if not the ablest, certainly one of ablest living experts. I said to Mr. Stone: 'Did you see patent?' He replied: 'Yes, I saw it, but I thought you crazy.' When I explained to Mr. Stone he said: 'Now, I why that is great,' and he understood how this energ transmitted.

"Now, gentlemen, we are coming to great results, bu must be prepared for a condition of paralysis for qui while. We are facing an enormous crisis such as the w has never seen before, and until the situation clears the thing we can do is to devise some scheme for overcoming submarines, and that is what I am doing now."